Seminar Paper

Urban Land-Use Efficiency

A Stable Metric for SDG 11.3.1 and Its Economic Drivers

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Abstract

The official SDG 11.3.1 indicator, the ratio of land consumption rate to population growth rate (LCRPGR), is unstable: it divides by a growth rate that approaches zero and mixes an arithmetic numerator with a logarithmic denominator. We propose the Built-up per Capita Rate (BpCR), the log change in built-up area per person, which equals the difference of two log growth rates and is therefore well-defined for every country-period. Using a harmonised global panel of 193 UN member states (1985–2020) built from GHSL built-up and population grids under the Degree of Urbanisation, we (i) document where LCRPGR breaks down, (ii) show BpCR reproduces the policy signal without the instability, and (iii) estimate its economic drivers with two-way fixed-effects and dynamic-panel GMM models.

Keywords: SDG 11.3.1, Urban land-use efficiency, Built-up per capita rate (BpCR), LCRPGR, Dynamic panel GMM, GHSL / Degree of Urbanisation

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1. Introduction

The world is urbanising rapidly: by the middle of this century close to two-thirds of humanity will live in cities, and almost all of the projected growth in built-up land will occur in developing countries (Gao and O'Neill, 2020). The scale of expansion is striking: every year the world's built-up area grows by roughly the size of the Netherlands (Seto et al., 2012). For development policy the central question is not whether cities expand, but how efficiently they do so, that is, how much additional land each new urban resident consumes. When built-up area grows in step with population, cities densify and remain compact; when it grows much faster than population, they sprawl. Sprawl carries real costs. Once farmland on the urban fringe is built over, it is effectively lost for good. Low-density, spread-out development makes people more dependent on cars, because homes, jobs, and services lie far apart. It also makes public infrastructure more expensive per resident, since the same roads, pipes, and power and transport networks must reach a smaller number of people over a larger area. And by encouraging longer car journeys and land-hungry construction, sprawl tends to raise greenhouse-gas emissions per person. Measuring land-use efficiency well is therefore a precondition for monitoring sustainable urbanisation.

The United Nations monitors this through target 11.3 of its Sustainable Development Goals (SDGs), whose official indicator 11.3.1 is the ratio of the Land Consumption Rate to the Population Growth Rate (LCRPGR) (UN-Habitat, 2025). Because LCRPGR is the agreed global yardstick, its statistical properties are not a technical detail: they determine how every country's progress is judged and compared. Yet the indicator is fragile by construction. It divides an arithmetic land-consumption rate by a logarithmic population-growth rate, so it is internally inconsistent; and because the population-growth rate sits in the denominator, the ratio diverges towards plus or minus infinity and changes sign whenever population is roughly stable. In exactly the countries where land-use efficiency matters most for policy, the official metric becomes uninformative.

This paper makes three contributions. First, it documents, on a harmonised global dataset, that LCRPGR is statistically unstable and that its instability is concentrated where population growth is near zero. Second, it proposes a simple, log-based alternative, the Built-up per Capita Rate (BpCR), defined as the logarithmic growth rate of built-up area per capita. BpCR is the difference of two logarithmic growth rates, which makes it finite for every country-period, symmetric between densification and sprawl, additively decomposable, and directly interpretable; it reproduces the policy signal of LCRPGR where the latter is well-behaved, and so complements rather than replaces the official indicator. Third, using the stable metric, the paper estimates the economic and structural drivers of per-capita urban land use on a panel of 193 United Nations member states observed at five-year intervals between 1985 and 2020.

The empirical analysis builds the panel from satellite-based settlement data under the global Degree of Urbanisation standard, so that built-up area and population are measured consistently. To address the well-known downward bias of the lagged dependent variable in dynamic fixed-effects models, the persistence of per-capita land use is estimated with Arellano-Bond and Blundell-Bond generalised method of moments (GMM) estimators. Three results stand out. Sprawl is the global norm rather than the exception: built-up area per capita rises in a clear majority of country-periods. Per-capita land use is strongly path-dependent, with about three-fifths of one period's value carried into the next. And within countries, denser cities, net in-migration, and higher income are all associated with less land consumed per resident, indicating that, over time, economic growth is accompanied by densification rather than sprawl.

The remainder of the paper is organised as follows. Section 2 reviews the indicator and the related literature. Section 3 sets out the structural limitations of LCRPGR, and Section 4 introduces BpCR. Section 5 describes the data and Section 6 presents descriptive evidence. Section 7 details the empirical strategy and Section 8 reports the results, including robustness checks. Section 9 discusses the findings and Section 10 concludes.

2. Literature review

2.1. Monitoring land-use efficiency: SDG 11.3.1

Target 11.3 of the Sustainable Development Goals commits countries to "enhance inclusive and sustainable urbanization." Its headline indicator, 11.3.1, is custodian-monitored by UN-Habitat and defined as the ratio of the Land Consumption Rate to the Population Growth Rate (LCRPGR), with an accompanying set of secondary indicators such as built-up area per capita (UN-Habitat, 2025). The intuition is simple and appealing: a value near one means that built-up land and population grow at the same pace, so the city neither densifies nor sprawls; a value above one signals that land is being consumed faster than population grows. Because the indicator is reported globally and used to benchmark countries, a large applied literature has grown up around how to compute it consistently from Earth-observation data (Corbane et al., 2017; Schiavina et al., 2023).

2.2. Drivers of urban land expansion

A first strand of research asks why cities expand and what makes them sprawl. Global remote-sensing evidence shows that built-up area has grown faster than urban population almost everywhere over recent decades, so that land consumed per resident has been rising worldwide (Seto et al., 2011, 2012). The same literature consistently identifies economic growth as a central driver: across countries, richer places tend to have expanded their urban footprint more, a pattern echoed in city-level panel studies that tie sprawl to rising income and population (Oueslati et al., 2015). Forward-looking simulations project that this expansion will continue through the twenty-first century, with most new urban land added in today's developing regions (Gao and O'Neill, 2020). This cross-country, mostly descriptive evidence motivates the drivers we examine, namely income, population pressure through migration, and the density of existing settlement, but it leaves open whether the same relationships hold within countries over time.

2.3. Measuring the indicator: data sensitivity and formulation critiques

A second, more recent strand turns the lens on the indicator itself. Two concerns recur. The first is data sensitivity: estimates of land-use efficiency depend heavily on which built-up and population datasets are used, so that a country's measured performance can change materially with the input data, which complicates cross-country comparison and calls for harmonised, transparent inputs (Lu and Weng, 2026; Zhong et al., 2025). The second is the formulation of LCRPGR itself. Nicolau et al. (2019) show, for the Portuguese case, that the official ratio is sensitive to how its components are defined and that logarithmic, efficiency-based formulations behave more reliably than the raw ratio. These critiques establish that the choice of metric, not only the choice of data, can change the conclusions drawn about sustainable urbanisation.

2.4. This paper

This paper connects the two strands. We take the measurement critique seriously by replacing the unstable LCRPGR with a stable, logarithmic metric, and we then use that metric to revisit the drivers question with a research design suited to panel data. Relative to the existing literature, the contribution is threefold: a metric (BpCR) that is well-defined for every country-period; a within-country, dynamic research design that corrects the bias of the lagged dependent variable; and global coverage of 193 United Nations member states observed at five-year intervals from 1985 to 2020.

3. Measurement framework

This section sets out the conceptual core of the paper. We first define the official SDG 11.3.1 indicator and show why it is statistically unstable, and we then introduce the stable, log-based alternative used in the empirical analysis.

3.1. *The SDG 11.3.1 indicator and its limitations*

Following the UN-Habitat metadata, the indicator divides an arithmetic land consumption rate by a logarithmic population growth rate:

$$\mathrm{LCRPGR} = \frac{\mathrm{LCR}}{\mathrm{PGR}}, \qquad \mathrm{LCR} = \frac{V_t - V_{t-z}}{V_{t-z}} \cdot \frac{1}{z}, \qquad \mathrm{PGR} = \frac{\ln(P_t/P_{t-z})}{z},$$

where V is built-up area, P population, and z the period length. Two structural problems follow. **(1) Division by a near-zero denominator.** When population is roughly stable ($\mathrm{PGR} \approx 0$) the ratio diverges towards $\pm\infty$ and flips sign, so small, ordinary demographic differences produce wild, uninterpretable values. **(2) Arithmetic-versus-logarithmic asymmetry.** The numerator is an arithmetic growth rate while the denominator is logarithmic, so the ratio is not symmetric and does not aggregate cleanly over time (Nicolau et al., 2019). The result is a metric whose tails are dominated by an arithmetic artefact rather than by real land-use behaviour.

3.2. *A stable metric: the Built-up per Capita Rate (BpCR)*

The UN-Habitat metadata itself lists built-up area per capita (V/P) as a secondary indicator. We build on it and measure its **log growth**:

$$\mathrm{BpCR}_{i,t} = \frac{1}{z} \ln\left(\frac{V_t/P_t}{V_{t-z}/P_{t-z}}\right) = \mathrm{LCR}_{\log} - \mathrm{PGR}_{\log}.$$

Because BpCR is the **difference of two logarithmic growth rates**, it is (i) finite and well-defined for every country-period (no division by ~0), (ii) symmetric (densification and sprawl are mirror images around zero), (iii) additively decomposable, and (iv) directly interpretable: $\mathrm{BpCR} > 0$ means sprawl (more land per resident), $\mathrm{BpCR} < 0$ densification, and $\mathrm{BpCR} = 0$ land growing exactly in step with population. BpCR reproduces the policy signal of LCRPGR where the latter is well-behaved, but without the instability; it complements rather than contradicts the official indicator.

4. Data

The analysis uses a single, **urban-scale** panel built under the EU/UN **Degree of Urbanisation** (urban = `GHS-SMOD` ≥ 21), so built-up and population are measured consistently from the same grids. Built-up area comes from **GHS-BUILT-S** and population from **GHS-POP** (GHSL R2023A) (Pesaresi et al., 2024; Schiavina et al., 2023), aggregated to national totals over **GAUL 2024** administrative units. Economic controls are GDP per capita (constant 2020 US$, UN SNAAMA) and net migration (UN DESA WPP 2024) (UN SNAAMA, 2025; UN DESA WPP, 2024); region and development group come from the UN M49 classification. The panel covers **193 UN member states** at 5-year epochs, **1985–2020**.

Table 1: Estimation sample

Quantity	Value
Countries	193
Country-period observations	2116

5. Descriptive statistics

Across 1985–2020 a clear majority of countries record $\mathrm{BpCR} > 0$: built-up area is expanding faster than urban population for most of the world, i.e. sprawl is the global norm rather than the exception. The contrast with LCRPGR is instructive – LCRPGR's distribution is dominated by extreme tail values concentrated exactly where population growth is near zero, whereas BpCR is well-behaved across the whole sample. Regional trajectories show convergence towards balance in some fast-urbanising economies while others continue to sprawl.

Table 2: Hypotheses

	Hypothesis	Expected
H1	BpCR captures land-use efficiency more reliably than LCRPGR	better fit
H2	Higher urban density is associated with less land per capita	–
H3	Higher income is associated with less land per capita	–
H4	Net in-migration densifies cities	–
H5	Per-capita land use is path-dependent	+

Table 3: Final dynamic-panel model: FE stepwise build-up, Arellano-Bond (difference) and Blundell-Bond (system) GMM. Coefficient with significance stars on top, clustered/robust standard error in parentheses below. *** p<0.01, ** p<0.05, * p<0.1.

	Base	+ Density	+ Net Migr.	+ GDP	+ Urban %	Arellano-Bond	Blundell-Bond
BpCR[t−1]	+0.2535***	+0.2346***	+0.2346***	+0.2286***	+0.2287***	+0.6057***	+0.5908***
	(0.0578)	(0.0582)	(0.0585)	(0.0583)	(0.0583)	(0.1081)	(0.0623)
ln(Urban Density)		-0.0147**	-0.0146**	-0.0169**	-0.0165**	-0.0523***	-0.0021**
		(0.0064)	(0.0065)	(0.0065)	(0.0065)	(0.0142)	(0.0010)
Net Migration (% Pop)			-0.0008**	-0.0008**	-0.0008**	-0.0005*	-0.0010***
			(0.0003)	(0.0003)	(0.0003)	(0.0003)	(0.0002)
ln(GDP per capita)				-0.0068**	-0.0068**	-0.0138**	-0.0004
				(0.0028)	(0.0028)	(0.0060)	(0.0003)
Urban Pop. Share					-0.0031	-0.0310	+0.0079*
					(0.0225)	(0.0520)	(0.0045)
Observations	1,499	1,499	1,499	1,499	1,499	1,351	2,895
Within R^2	0.076	0.091	0.108	0.125	0.125	–	–
Country FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Period FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sargan (p)	–	–	–	–	–	0.44	0.14
AR(2) (p)	–	–	–	–	–	0.71	0.47

6. Empirical strategy

Research question. What are the economic drivers of per-capita urban land use, and how persistent is it over time? We test five hypotheses (Table 2).

We run **one specification** with each metric as the dependent variable, so the comparison is like-for-like:

$$\text{Metric}_{i,t} = \rho\,\text{Metric}_{i,t-1} + \beta_1 \ln(\text{UrbDens})_{i,t} + \beta_2\,\text{UrbShare}_{i,t} + \beta_3 \ln(\text{GDPpc})_{i,t} + \beta_4\,\text{NetMigr}_{i,t} + \mu_i + \delta_t + \varepsilon_{i,t},$$

with country fixed effects μ_i, period fixed effects δ_t, and standard errors clustered by country. We proceed in two stages. **(1) Static** (dropping the lagged dependent variable) to isolate the role of the *metric* – LCRPGR versus BpCR. **(2) Dynamic** (adding the lagged DV) to capture path-dependence. Because the lagged-DV coefficient is **downward-biased under fixed effects (Nickell bias)**, we correct it with **Arellano-Bond difference GMM** and **Blundell-Bond system GMM**, instrumenting with deeper lags. Instrument validity is assessed with the **Sargan/Hansen** over-identification test and the **AR(2)** serial-correlation test (both $p > 0.10$ indicating validity).

7. Results

The metric matters. On identical data, the static LCRPGR specification is contaminated by its unstable tails, while the BpCR specification is well-behaved; the two diverge precisely where PGR is small, confirming that LCRPGR's extremes are an arithmetic artefact rather than a behavioural signal.

The final dynamic model. Table 3 reports the preferred specification: a fixed-effects stepwise build-up alongside the Arellano-Bond and Blundell-Bond GMM estimates.

In the Arellano-Bond specification the lagged term is large and significant ($\rho \approx 0.61$): roughly 60% of last period's BpCR persists, and the GMM ρ is about three times the FE estimate, exactly as the Nickell bias predicts, confirming **strong path-dependence** in per-capita land use. Among the drivers, **urban density** enters negative (a compact-city effect: denser cities consume less new land per resident), **net migration** negative (in-migrants pack into the existing stock faster than built-up grows), and **GDP per capita** negative – within countries, getting richer is associated with **densification, not sprawl**. The level of urbanisation (urban population share) is insignificant once the rest is controlled, suggesting that urban **form**, not stage, drives land use.

8. Robustness

The findings are stable across a 2000–2020 subsample, a restriction to shrinking-population countries, and alternative instrument depths; the Sargan and AR(2) diagnostics pass throughout, supporting instrument validity.

9. Discussion

The within-country result that income is associated with densification may appear to contradict the cross-sectional literature in which richer countries have historically sprawled. The two are complementary: *across* countries, higher income has coincided with more sprawl historically, but *within* a country, rising income over time is associated with denser, not more sprawling, urban land use. Measuring this distinction requires a metric that is stable within country-period – which BpCR provides and LCRPGR does not.

10. Conclusion

The official SDG 11.3.1 indicator is statistically unstable by construction. The Built-up per Capita Rate is a drop-in, log-based alternative that is finite, symmetric, and interpretable for every country-period, and it reproduces the policy signal without the instability. Estimated dynamically on a global panel, per-capita urban land use is strongly path-dependent and, within countries, declines with density, in-migration, and income. The contribution is as much **methodological** – a stable metric and a bias-corrected dynamic estimator – as substantive, and it complements rather than overturns the existing literature.

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